

Johny Yong Jun Siang

Johnnyong10@gmail.com | +65 93726829 | <https://johnnyong.github.io/DeveloperPortfolio/#/> | www.linkedin.com/in/johnnyong

SUMMARY

Software Developer and **B.Sc. Interactive Media & Game Development** student with over 5 years of experience in C++, **Python**, and C# across diverse platforms. Beyond technical development, I have a proven track record of **leading cross-functional teams**, managing product roadmaps, and authoring technical documentation for complex projects like custom game engines. I offer a unique blend of **software engineering** expertise and **project coordination** skills, with a professional and adaptable approach to troubleshooting. I am eager to leverage my background in streamlining development workflows to contribute to **Corp IT** objectives as an **IT Analyst** during the **Garena Integrated Work Study Programme (IWSP)**.

NOTABLE PROJECTS

Battlemania

- **Challenged myself to master 2D fighting game mechanics**, independently researching and implementing frame-data logic and precise melee hitboxes.
- **Having learnt Forge Networking, I expanded my networking expertise** by experimenting with the **Photon Networking** package to solve latency issues in high-speed combat scenarios.
- **Adopted a "systems-first" mindset**, focusing on building a robust underlying engine for combat and movement before integrating placeholder art assets.
- **Managed the full development lifecycle** as a solo creator, from initial concept to a functional multiplayer prototype.

Gravity

- Led a 9-member team to develop a physics-driven top-down survival game built on a **custom C++ engine**, managing the product roadmap and technical milestones.
- Designed the engine's **Entity Component System (ECS)**, including a Game Object Factory and a Tag/Layer Manager to optimize object handling and performance.
- Authored comprehensive **Technical Design Documentation** using **MkDocs**, maintaining a live version via GitHub Pages for team alignment and version control.

AzureVania

- **Spearheaded the multiplayer architecture** by integrating the Forge Networking package which we are exposed to for the first time, ensuring a seamless data sync between players in a 2D Metroidvania environment.
- **Owned the end-to-end level design process**, creating an interconnected world that balanced player progression with technical performance constraints.
- **Designed and refined Enemy AI behaviors**, utilizing logic-based decision-making to create challenging and responsive combat encounters.
- **Facilitated team synchronization**, bridging the gap between design vision and technical implementation to deliver a cohesive final prototype.

EDUCATION

B.Sc.(Hons), Interactive Media & Game Development
Singapore Institute of Technology / DigiPen Institute of Technology
Diploma in Game Development and Technology
Nanyang Polytechnic

~August 2023 - April 2027

~April 2018 - April 2021

SKILLS

Programming & Scripting: C, C++, C#, Python (Intermediate), Java, HTML, Lua, JavaScript.

Networking & Backend: Photon Networking, Forge Networking, Firebase, Azure DevOps, JSON

Software & Tools: Visual Studio 2022, Visual Studio Code, GitHub, Android Studio, Flutter, MkDocs

Project Management & Design: Technical Design Documentation, Cross-functional Team Leadership, UI/UX Design, Agile Workflow

Spoken Languages: English (Fluent), Mandarin (Fluent)

ACHIEVEMENTS

Edusave Good Progress Award

Nanyang Polytechnic

Commanding Officer's Coin

Singapore Armed Forces

EXPERIENCE

Mentor, Singapore Institute of Technology Community Challenge

Singapore Institute of Technology - Part Time

~October 2025 - October 2025

- I volunteered to mentor teams in using **VittaScience AI** posture & image recognition tools for prototype development.
- Supported the students across 5 days on their programming workflows, feature planning, UI/UX design decisions, and technical troubleshooting.

Auto Technician

Singapore Armed Forces - National Service

~May 2021 - August 2023

- Maintained and repaired military vehicle systems as an auto technician
- Collaborated within a technical team under strict deadlines and safety protocols
- Developed a storage management mobile app during National Service using Azure DevOps & Flutter for development and Firebase for real-time tracking of resource usage.

Software Engineer

YJP Surveyors Pte Ltd - Internship

~June 2020 - August 2020

- Contributed to a HoloLens mixed reality project built in Unity, enhancing visualisation of underground pipes and connections normally hidden by walls or flooring.
- Implemented **voice recognition** features to improve user interaction and accessibility.
- Gained hands-on experience with point cloud data and **LiDar technology** to support spatial mapping and alignment.
- Delivered a fully functional prototype that was commended for its effectiveness and innovation.

CERTIFICATIONS

Playstation Game Development Training Completion

- Successfully completed industry recognised training in Playstation workflows by utilising the Playstation DevKit.

AI For Industry: Literacy in AI

- Gained foundational knowledge of artificial intelligence concepts and their industry applications.